SIEU TRAN

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RESEARCH INTERESTS

Structured 3D scene representations and world models for embodied agents in dynamic and partially observable environments. I am particularly interested in object-centric and relational representations that maintain persistent state through occlusion and change, capture geometry and physical relationships between objects, and support prediction, reasoning, and long-horizon decision making.

EDUCATION

University of Arkansas (UARK)
Department of Electrical Engineering and Computer Science
Master of Science in Computer Science CGPA: 4.00/4.00

Vietnam National University Ho Chi Minh City (VNU-HCMC)
University of Information Technology (UIT)
Bachelor of Science in Computer Science GPA: 9.24/10.0 Rank: 1st – Valedictorian

EXPERIENCE

University of Arkansas – AICV Lab 2026 – Present

- Proposed a fully self-supervised object-centric video framework that separates per-frame appearance from temporally persistent identity into two slot states, reconciling reconstruction quality with temporal consistency; **+9.7 FG-ARI** over the strongest baseline on YouTube-VIS, also evaluated on MOVi-C and MOVi-D (*DSSA*, ACM MM 2026, accepted, co-first author).
- Designed a self-supervised slot-lifecycle mechanism that learns per-slot temporal activation **without visibility supervision**, preserving latent state for absent or fully occluded objects while reducing state drift and decoder interference (*TSA*, under review, co-first author).
- Designed the persistent semantic-graph state and proposed executable spatial predicates for subtask-completion checking in a hierarchical **vision-language-action** framework for non-Markovian long-horizon robot manipulation; **81.7% vs. 56.7%** average success over the strongest history-enabled baseline on three real-robot UR10e tasks (*CodeGraphVLP*, under review, co-first author).
- Contributed to geometry-aware multi-view understanding for driving MLLMs and to a benchmark for spatiotemporal reasoning in vision-language models (*OmniSpace* and *DriveSpatial*, NeurIPS 2026).

Day One Company – AI Lab 2024 – 2025

- Proposed a **self-supervised domain adaptation** pipeline enabling OCR models to adapt to new customer domains without additional annotation, reducing onboarding cost by 45%.
- Built information extraction systems that generalized across unseen receipt layouts, increasing end-to-end automation from 60% to 90% and reaching 99% accuracy on fraudulent-field detection.
- Led research, development, deployment, and maintenance of production AI systems, including MLOps pipelines and model monitoring.

University of Information Technology – AI Lab 2021 – 2024

- Developed region-aware pose representations and **signer-invariant** learning strategies for generalization across individuals in signer-independent sign language recognition (2 papers, ICCV Workshop 2025; 2nd place, MSLR Challenge).
- Built multimodal retrieval systems combining vision-language representations (CLIP, BLIP), OCR, ASR, and temporal modeling for search over 300+ hours of video (SOICT 2023, first author).
- Developed object-detection systems for severe class imbalance and adverse visual conditions in real-world traffic footage (CVPR Workshop 2024; 1st place, Track 5, 8th AI City Challenge).

PUBLICATIONS

[1] **Sieu Tran**[*], Duc Nguyen Minh[*], Hao Vo, Khoa Vo, and Ngan Le. "Dual-State Slot Attention: Decoupling Appearance and Identity for Video Object-Centric Learning." *ACM International Conference on Multimedia (ACM MM)*, 2026. **Accepted.**

[2] Duc Nguyen Minh[*], **Sieu Tran**[*], Hao Vo, Khoa Vo, Duy Minh Ho Nguyen, Nghi D. Q. Bui, Anh Nguyen, Long Mai, and Ngan Le. "TSA: Temporal Slot Activation for Persistent Object-Centric Video Representation." *arXiv preprint*, 2026.

[3] Khoa Vo[*], **Sieu Tran**[*], Taisei Hanyu, Yuki Ikebe, Duy Nguyen, Nghi D. Q. Bui, Minh Vu, Anthony Gunderman, Chase Rainwater, Anh Nguyen, and Ngan Le. "CodeGraphVLP: Code-as-Planner Meets Semantic-Graph State for Non-Markovian Vision-Language-Action Models." *arXiv preprint*, 2026.

[4] Hao Vo, Phu Loc Nguyen, Khoa Vo, **Sieu Tran**, Duc Minh Nguyen, Ngo Xuan Cuong, Nghi D. Q. Bui, Anh Nguyen, Duy Minh Ho Nguyen, and Ngan Le. "OmniSpace: Efficient Geometry Awareness for Autonomous Vehicles MLLMs." *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. **Accepted.**

[5] Hao Vo, Khoa Vo, Phu Loc Nguyen, **Sieu Tran**, Duc Minh Nguyen, Ngo Xuan Cuong, Gladys Gawugah, Sreevenkata Anjani Tishita Godavarthi, Chase Rainwater, Nghi D. Q. Bui, Anh Nguyen, Duy Minh Ho Nguyen, and Ngan Le. "DRIVESPATIAL: A Benchmark for Spatiotemporal Intelligence in VLMs for Autonomous Driving." *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. **Accepted.**

[6] **Sieu Tran**, Duc Nguyen Minh, Truong Nguyen Thanh, Hao Vo, Thanh Duc Ngo, Tien Do, Khiem Le, and Duy-Dinh Le. "Region-Aware Pose Modeling and Permutation Decoding for Signer-Independent Sign Language Recognition." *IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2025.

[7] **Sieu Tran**, Duc Nguyen Minh, Truong Nguyen Thanh, Hao Vo, Thanh Duc Ngo, Tien Do, Khiem Le, and Duy-Dinh Le. "Generalizable Sign Language Recognition via Local Temporal Convolutions and Region-Aware Pose Encoding." *IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2025.

[8] Hao Vo, **Sieu Tran**, Duc Minh Nguyen, Thua Nguyen, Tien Do, Duy-Dinh Le, and Thanh Duc Ngo. "Robust Motorcycle Helmet Detection in Real-World Scenarios: Using Co-DETR and Minority Class Enhancement." *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2024.

[9] **Sieu Tran**, Duc Minh Nguyen, Triet Huynh Minh Nguyen, Danh Phuc Ngo, Thu Minh Nguyen, Hao Vo, Khiem Le, Tien Do, and Thanh Duc Ngo. "Diverse Search Methods and Multi-Modal Fusion for High-Performance Video Retrieval." *International Symposium on Information and Communication Technology (SOICT)*, 2023.

[*]Equal contribution.

SELECTED HONORS AND AWARDS

Graduate Assistantship Award, University of Arkansas	2026
Two Challenge Awards, Multimodal Sign Language Recognition, ICCV Workshop	2025
1st Place, Track 5, 8th AI City Challenge, CVPR Workshop	2024
Valedictorian, Bachelor of Science, UIT	2024
Second Prize, National Science and Technology Award for University Students, Vietnam	2024